



RAMON MAGSAYSAY MEMORIAL COLLEGES

Information Technology Education Program

1st SEMESTER: AY: 2025 - 2026



NAME: JOHN MICHAEL R. CASTILLO SCHEDULE: F-S 1:00PM – 4:00PM

SCORE: _____

SUBJECT: **WEB SYSTEMS AND TECHNOLOGIES** INSTRUCTOR: JIM JAMERO DATE: _____

LABORATORY EXERCISE 2 DATABASE DESIGN AND MIGRATION

Learning Objectives

- By the end of this laboratory exercise, students should be able to:
 - Design a database schema for a Learning Management System (LMS).
 - Identify and define core database tables and relationships.
 - Implement database migrations using CodeIgniter's migration feature.
 - Seed the database with sample data for testing purposes.
 - Use GitHub for version control and tracking database changes.

Prerequisite student experiences and knowledge

Before starting this exercise, students should have:

- ❖ Basic knowledge of relational database concepts.
- ❖ Familiarity with MySQL database creation and management.
- ❖ Understanding of CodeIgniter framework basics.
- ❖ Experience using Git and GitHub for project version control.
- ❖ Ability to run and configure a local web server (XAMPP/WAMP/LAMP).

Background

This laboratory focuses on creating the database backbone for the LMS project. The process includes identifying core tables, defining their structure, setting up migrations for automated schema creation, and seeding data for testing. Using migrations ensures version-controlled database changes that can be easily shared among team members through GitHub.

Materials/Resources

- Personal Computer with Internet Access
- XAMPP/WAMP/LAMP server installed
- CodeIgniter Framework (latest version)
- Bootstrap (via CDN or downloaded files)
- Visual Studio Code or any code editor
- Git and GitHub Account
- Web Browser (Chrome, Firefox, etc.)

Laboratory Activity

Step 1: Identify Core Tables

List the primary tables for the LMS:

1. users – stores student, instructor, and admin information.
2. courses – contains course details.
3. enrollments – manages student enrollment in courses.
4. lessons – stores lesson content linked to courses.
5. quizzes – contains quiz questions linked to lessons.



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6. submissions – stores quiz submissions and results.

Step 2: Create Database

1. Open phpMyAdmin.
2. Create a new database named: lms_lastname
3. Do not create tables manually — migrations will handle this.

Step 3: Enable CodeIgniter Migrations

1. In `application/config/migration.php`, set:
 \$config['migration_enabled'] = TRUE;
 \$config['migration_type'] = 'sequential';
2. In `application/config/config.php`, set your database connection details.

Step 4: Create Migration Files

1. In the terminal, navigate to your project folder.
2. Create migration files for each table:
 php spark make:migration CreateUserTable
 php spark make:migration CreateCoursesTable
 php spark make:migration CreateEnrollmentsTable
 php spark make:migration CreateLessonsTable
 php spark make:migration CreateQuizzesTable
 php spark make:migration CreateSubmissionsTable
3. Define the schema in each migration file.

Step 5: Run Migrations

Run the following command:

- php spark migrate

Step 6: Seed Sample Data

1. Create a seeder: **php spark make:seeder UserSeeder**
2. Add sample users (students, instructors, admin).
3. Run the seeder: **php spark db:seed UserSeeder**

Step 7: Push to GitHub

1. Stage and commit your migration and seeder files:
 git add .
 git commit -m "Added database migrations and seeders"
 git push

Output / Results

- ✓ Screenshot of successful migration in phpMyAdmin.
- ✓ Screenshot of seeded sample data in phpMyAdmin.
- ✓ Screenshot of GitHub repository with migration files.



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QUESTIONS:

1. Why are migrations important in database development?

Migrations are important because they help us manage the structure of the database in a clear and organized way. Instead of manually creating or editing tables, we use migrations to write changes in code. This makes it easier to track what was changed, who changed it, and when. It also allows us to roll back if we make a mistake or need to update something later.

2. How does using GitHub with migrations help in team projects?

Using GitHub with migrations helps teams work together smoothly. When each member pushes their migration files to GitHub, others can pull and apply the same changes to their own local databases. This avoids confusion and ensures everyone is working with the same database structure, which is especially helpful when many developers are working on the same project.

3. What is the advantage of seeding sample data during development?

Seeding sample data is helpful because it gives us fake or test data that we can use while building and testing our system. Instead of manually entering users or records every time, we can run a seeder and have the data ready instantly. This saves time and allows us to test features more efficiently, especially when dealing with forms, queries, or login systems.



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Output / Results

```
Windows PowerShell
You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right s
yntax to use near '' at line 1
at SYSTEMPATH\Database\MySql\Connection.php:327

PS C:\xampp\htdocs\ITE311-CASTILLO> php spark migrate

CodeIgniter v4.6.3 Command Line Tool - Server Time: 2025-08-14 06:58:06 UTC+00:00

Running all new migrations...
Running: (App) 2025-08-14-052911_App\Database\Migrations\CreateUsersTable
Running: (App) 2025-08-14-052912_App\Database\Migrations\CreateCoursesTable
Running: (App) 2025-08-14-052913_App\Database\Migrations\CreateEnrollmentsTable
Running: (App) 2025-08-14-052913_App\Database\Migrations\CreateLessonsTable
Running: (App) 2025-08-14-052913_App\Database\Migrations\CreateQuizzesTable
Running: (App) 2025-08-14-052925_App\Database\Migrations\CreateSubmissionsTable
Migrations complete.
PS C:\xampp\htdocs\ITE311-CASTILLO> php spark make:seeder UsersSeeder

CodeIgniter v4.6.3 Command Line Tool - Server Time: 2025-08-14 07:04:21 UTC+00:00

File created: APPPATH\Database\Seeds\UsersSeeder.php

PS C:\xampp\htdocs\ITE311-CASTILLO> php spark db:seed UsersSeeder

CodeIgniter v4.6.3 Command Line Tool - Server Time: 2025-08-14 07:08:00 UTC+00:00

Seeded: App\Database\Seeds\UsersSeeder
PS C:\xampp\htdocs\ITE311-CASTILLO> php spark db:seed UsersSeeder

CodeIgniter v4.6.3 Command Line Tool - Server Time: 2025-08-14 07:10:19 UTC+00:00
```

id	role	name	email	password	created_at	updated_at
1	admin	John Michael Castillo	jmykel1342@gmail.com	\$2y\$10\$0LNE3C9K4M0zBar1T1YW0E10XVb1ay2e1fYv...	NULL	NULL
2	student	Zaid Adlan	zai@gmail.com	\$2y\$10\$K4sOrBNBb1t8R5e4a3uTyme3acPLFXJv3CRW...	NULL	NULL

Commits

main

Commits on Aug 14, 2025

Laboratory 2: Setup DB with migration and seeding

jmykel0325 committed 3 minutes ago

b0bdb99



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Conclusion

This activity helped me understand how to use migrations and seeders in CodeIgniter. I learned how migrations make it easier to manage database changes, especially when working with a team. I also saw how seeding is useful for adding sample data for testing. Overall, this activity showed me how important it is to keep the database organized and how tools like GitHub help make group work easier and more efficient.





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